

# Activity trace — extending matched SHD training

ar071 · 19 July 2026 · Draft

## Checkpoint CP-001

This publication-to-mandate checkpoint has immutable private source SHA-256 prefix `4b08b55d459f`. Its checkpoint time is `2026-07-19 00:49:00.457 UTC`. It has no predecessor within this night. Hidden reasoning, tool payloads, injected environment material, credentials, private paths, and infrastructure identifiers are excluded; the one visible private local link is explicitly redacted below.

### Decision and action ledger

1. Pull request 51 was explicitly accepted and merged into `main` using merge commit `569926f`; no squash or rebase was used.
2. The accepted result motivates higher accuracy, but repeated official-test use was rejected as an optimization loop. The next outcome is held-out validation accuracy on `exp068`'s identical deterministic split.
3. The approved intervention changes only fixed training duration, from 40 to 80 epochs, in both matched cells. One seed remains appropriate for this exploratory mechanism check.
4. New identifiers are `ar070`, `ar071`, and `exp069`. At approval, `exp069` spend was 0 USD, no pod was active, and no implementation or result existed.

## Privacy-sanitized visible transcript

Assistant

2026-07-19 00:41:04.944 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`Exp068` is complete and stopped at the human-review gate. `PR 51` remains draft and unmerged.

Key evidence:

- PING: **45.72%** official-test accuracy
- COBA: **35.07%**
- Difference: **+10.64 percentage points**
- PING/COBA macro accuracy: 45.47% / 34.64%
- PING/COBA cross-entropy: 2.055 / 2.345
- PING E/I activity: 10.50 / 47.99 Hz
- COBA E activity: 28.44 Hz
- One seed; all 2,264 official-test samples evaluated after paired checkpoint freeze
- Zero skipped updates or non-finite batches
- Exact recorded compute spend: **2.4414148 USD**
- Active pods: **0**

Links included PR 51 and the rendered ar068, exp068, and ar069 pages. (PRIVATE LOCAL PATH REDACTED)

Validation passed: focused Ruff and ty, compilation, 20 artifact hashes, registered-result assertions, raster inspection, zero-stub 35-entry Demolab build, and GitHub preview. Repository-wide CI lint remains red solely from previously documented ty debt; the test job was still running at handoff.

Final commits: 7671ce4 and ae7a26a. Goal usage: 5,630 tokens over roughly 18 seconds of tracked goal execution.

## User

2026-07-19 00:44:59.911 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

merge into main, then make a new pr we gonna try get those acc numbers higher

## Assistant

2026-07-19 00:45:05.441 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll merge PR 51 into `main` at the publication gate first, preserving its merge history. Then I'll synchronize `main` and open a fresh research branch/PR for the accuracy-improvement night; I'll keep that new experiment at the mandate-design gate until its scientific setup is explicit. I'm using the GitHub workflow guidance because this changes PR and branch state.

## Assistant

2026-07-19 00:45:15.365 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is mergeable and the preview is green. Its lint failure is still only the previously documented repository-wide type debt, with no exp068 diagnostic; the long test job remains in progress. Your explicit merge decision resolves the publication gate, so I'm marking the draft ready and merging it now without squashing.

## Assistant

2026-07-19 00:45:28.680 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is merged into `main` as merge commit 569926f. This closes the exp068 publication gate. I'm synchronizing that accepted state and checking the autoresearch registration rules and next available IDs before creating the new accuracy-improvement PR.

## Assistant

2026-07-19 00:46:05.090 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is merged into `main` as merge commit `569926f`.

Before opening the next PR, the autoresearch mandate gate requires a locked scientific design. I recommend this clean next step:

- One seed, same deterministic 7,340/816 SHD development split.
- Keep the official test sealed; repeated use would turn it into an optimization target.
- Train matched COBA and PING for 80 epochs instead of 40.
- Preserve every `exp068` architecture, optimizer, input, and readout setting.
- Select checkpoints using validation accuracy, then cross-entropy, then earlier epoch.
- Primary criterion: PING improves at least 3 percentage points over its `exp068` 43.50% validation baseline.
- Secondary: whether COBA improves over 40.81%, whether PING remains ahead, and activity/runtime diagnostics.
- One local smoke, then two concurrent pods, 10 USD ceiling, no `tools/snn` edits.
- New IDs: `ar070`, `ar071`, and `exp069`.
- Only after approval: lock the mandate on `main`, create the night branch, and open the new draft PR.

Reply `go` to approve this mandate.

**User**

2026-07-19 00:48:51.919 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`go`

**Assistant**

2026-07-19 00:49:00.457 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Mandate approved. I'm locking `ar070`, `ar071`, and `exp069` on `main` now with the 80-epoch, validation-only exploratory contract. This approval starts no compute: after the anchor is pushed, I'll create the dedicated night branch and new draft PR from that exact commit.

## Checkpoint CP-002

This branch-and-PR checkpoint follows CP-001. Its immutable private source has SHA-256 prefix `77b86b9fea8d`, and its checkpoint time is `2026-07-19 00:53:19.217 UTC`.

### Action ledger

1. The approved mandate was committed and pushed to `main` at `3c440aa`.
2. The night branch was created from that exact anchor and its first Record checkpoint was committed at `8229b83`.
3. Draft pull request 52 is open. No experiment runner, smoke, pod, result, or cost exists; `exp069` spend remains 0 USD.

### Visible messages added in CP-002

**Assistant**

2026-07-19 00:50:48.822 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate itself builds, but the initial activity-trace rendering found an unclosed Typst delimiter in a verbatim message containing bracket and currency punctuation. I'm escaping only that transcript presentation; the approved scientific contract remains unchanged.

### Assistant

2026-07-19 00:52:07.816 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate is now preregistered on `main` at `3c440aa`; `exp069` spend is 0 USD and no compute has started. I'm creating `night/spiking-heidelberg-digits/ar070` from that exact anchor, adding the branch-start checkpoint, then pushing and opening the new draft PR.

### Assistant

2026-07-19 00:53:19.217 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The new draft PR is open as 52 from the exact locked anchor. I'm adding its timestamped branch/PR checkpoint to `ar071` and pushing that final setup commit; `exp069` remains unrun with 0 USD spend and no active pods.

## Checkpoint CP-003

This implementation-and-smoke checkpoint follows CP-002. Its immutable private source has SHA-256 prefix `8b2ba5ea34fb`, and its checkpoint time is `2026-07-19 01:39:56.429 UTC`. One private repository path in a visible handoff prompt is redacted; injected environment/context messages, hidden reasoning, and tool payloads are excluded.

### Decision and action ledger

1. `Exp069` was implemented through `experiments/exp069.py` without editing `tools/snn`. The official-test route present in `exp068` was removed rather than inherited, and matched rasters were redirected to validation data.
2. Focused Ruff, type, and compilation checks passed.
3. The registered local smoke passed both cells. Split hashes match `exp068`; losses and rates were finite; skipped and non-finite update counters were zero; COBA E and PING E/I populations were active.
4. No cloud compute had started and spend remained 0 USD.

### Visible messages added in CP-003

### Assistant

2026-07-19 00:54:21.949 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Done.

- PR 51 merged into `main`: `569926f`

- Exp069 mandate locked on main: 3c440aa
- New branch: night/spiking-heidelberg-digits/ar070
- New draft PR: PR 52
- Activity checkpoint pushed: 08edd92
- Demolab build: 37 entries, zero stubs
- Current RunPod spend: \$0
- Active pods: 0
- No experiment or official-test access has started yet.

**User**

2026-07-19 01:14:49.989 UTC

Role: user · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

do it

**User**

2026-07-19 01:15:12.669 UTC

Role: user · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

give me the goal to give you so I can compact convo

**Assistant**

2026-07-19 01:15:30.248 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Use this after /compact:

/goal Continue and complete the locked exp069 autoresearch night on PR #52.

Repository and state:

- Work in (PRIVATE LOCAL PATH REDACTED).
- Current branch: night/spiking-heidelberg-digits/ar070.
- PR 51/exp068 was accepted and merged into main as 569926f.
- The exp069 mandate is locked in writings/ar070.typ on main at 3c440aa.
- Draft PR 52: <https://github.com/eoinmurray/pinglab/pull/52>
- Latest night checkpoint: 08edd92.
- Activity trace: writings/ar071.typ.
- Experiment/report IDs: exp069 and writings/exp069.typ.
- No exp069 compute has started; recorded spend is \$0 and there should be zero active pods.

Execute the registered design exactly:

- Test whether extending matched training from 40 to 80 epochs raises PING's selected validation accuracy by at least 3 percentage points over exp068's 43.50% baseline.
- One seed: 42.
- Reuse exp068's exact deterministic 7,340-training/816-validation split and verify its index hashes.
- Preserve every exp068 architecture, optimizer, input, readout, ordering, and preprocessing setting.
- The only scientific change is 80 training epochs instead of 40.
- COBA: inhibitory loop disabled and no voltage-gradient dampening.
- PING: inhibitory loop enabled and dampening 1000.

- Use the registered checkpoint rule: highest validation accuracy, then lower validation cross-entropy, then earlier epoch.
- The official SHD test is forbidden: do not stage, load, inspect, evaluate, or report it.
- No architecture-specific tuning, rescue, or post-result design changes.
- Do not edit tools/snn without asking first.

Execution protocol:

1. Run the registered local 128-training/128-validation, two-epoch smoke first.
2. Require finite optimization, active populations, matched partitions, and no persistent skipped updates.
3. If the gate passes, run one full 80-epoch cell per architecture.
4. RunPod authorization: at most two concurrent pods, one per architecture, with a hard total ceiling of \$10 and four-hour per-pod backstops.
5. Reap pods after completion or failure and verify zero remain.
6. Do not run the full test suite on the 4 GB host; use focused Ruff, ty, compilation/integrity checks, and demolab build.

Deliverables:

- Implement through experiments/exp069.py using existing CLI capabilities.
- Produce artifacts/data/exp069 with provenance, split hashes, numbers.json, reproducer, learning/activity curves, runtime and memory diagnostics, skipped/non-finite counters, and matched validation input/E/I rasters.
- Create writings/exp069.typ as a cold-readable artifact-driven report.
- Update the timestamped Record and Digest in writings/ar070.typ.
- Continue writings/ar071.typ with verbatim chronological user-visible messages and timestamped, hash-linked checkpoints. Exclude hidden reasoning and tool payloads; sanitize credentials, tokens, SSH material, IPs, private paths, environment values, pod identifiers, and sensitive infrastructure. Maintain immutable private raw transcript checkpoints outside the repository and record their SHA-256 prefix hashes.
- Add timestamped PR 52 comments at meaningful scientific and compute milestones.
- Make focused commits for implementation, smoke, dispatch, results, failures, and publication; preserve killed attempts.
- Build successfully, push the existing branch, and update PR 52
- Do not create another branch or PR, modify main, merge PR 52, or use the official test.
- Stop and ask before changing the registered scientific design, editing tools/snn, exceeding the compute budget, or taking another action requiring authority.
- Finish at the human publication gate with a concise evidence summary, exact compute spend, validation performed, rendered ar070/ar071/exp069 links, and PR 52 Leave it draft and unmerged.

**Assistant**

2026-07-19 01:33:02.681 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm resuming exp069 from the locked branch state. I'll first verify the Demolab runbook, worktree, PR/pod state, and mandate, then implement and smoke-test before any authorized compute dispatch. I'll also checkpoint this resumed visible thread into the sanitized activity log before publication.

**Assistant**

2026-07-19 01:33:14.646 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The branch is clean at checkpoint **08edd92**, tracks the intended remote, and the provider reports zero active pods, so the registered \$0 starting state is intact. I'm now reading the active mandate/runbook and comparing exp068's runner/artifacts so exp069 changes only the epoch count and test access is structurally excluded.

### Assistant

2026-07-19 01:33:59.312 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The reuse audit found the key safety issue: exp068 contains a post-freeze official-test path, so exp069 cannot merely change **40** to **80**. I'm deriving the runner while deleting every download/evaluation route and changing rasters to fixed validation examples; the training recipe and selection logic remain identical.

### Assistant

2026-07-19 01:35:43.759 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Focused Ruff, type, and compilation checks pass. The runner now pins the exact exp068 split hashes, sets only 80 epochs as the scientific change, uses a four-hour pod backstop, and has no official-test URL/import/evaluator. I'm starting the mandatory 128/128, two-epoch local smoke now; no cloud spend is involved.

### Assistant

2026-07-19 01:39:56.429 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The registered smoke gate passed exactly: both cells were finite, all update counters stayed clean, COBA E activity was 23.93 Hz, and PING E/I activity was 2.95/14.56 Hz. The split hashes exactly match exp068. I'm checkpointing the implementation and smoke evidence—plus the resumed visible transcript—before dispatching either pod.

## Checkpoint CP-004

This compute-dispatch checkpoint follows commit **b944ebf**. Its immutable private source has SHA-256 prefix **2006f03650d5**, and its checkpoint time is **2026-07-19 01:43:23.094 UTC**.

Provider pod identifiers and sensitive infrastructure details are excluded.

### Compute ledger

1. Exactly two single-GPU pods were dispatched, one per architecture, at 0.99 USD/hour each. Four-hour backstops cap exposure at 7.92 USD.
2. A transient PING capacity miss was recovered by the existing bounded provisioning retry. The resulting active fleet was exactly two.
3. Both cells are pinned to **b944ebf**; official-test access remains forbidden.

## Visible messages added in CP-004

### Assistant

2026-07-19 01:42:35.132 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Smoke checkpoint **b944ebf** is pushed and the rendered trace/night PDFs were rebuilt. The gate authorizes full compute, so I'm posting the timestamped smoke milestone to PR 52, then I'll dry-run the exact two-cell dispatch before using `--live`. I'm applying the GitHub publishing workflow because these PR writes are explicitly part of the mandate.

### Assistant

2026-07-19 01:42:46.629 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The dry-run resolves to exactly two 5090 pods—one COBA and one PING—with no extra jobs. The smoke checkpoint is pinned and the projected worst-case backstop cost remains below the authorized \$10 ceiling. I'm dispatching those two pods now; immediately afterward I'll record their creation time and verify the fleet count is exactly two.

### Assistant

2026-07-19 01:43:23.094 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods are now running at \$0.99/hour each, pinned to **b944ebf**; the first PING capacity request briefly found no stock, then the registered retry succeeded without creating an extra pod. Fleet count is exactly two, so the maximum four-hour exposure is \$7.92—inside the \$10 ceiling. I'm recording this dispatch milestone while the pods train.

## Checkpoint CP-005

This collection-and-result checkpoint follows CP-004. Its immutable private source has SHA-256 prefix **77c050cd9f2b**, and its checkpoint time is **2026-07-19 04:12:10.443 UTC**. Provider identifiers, private paths, injected context, tool payloads, and hidden reasoning are excluded.

### Scientific and compute ledger

1. PING completed eighty epochs and selected epoch 79 at 48.04% validation accuracy, +4.54 points over its registered 43.50% baseline. The +3-point primary criterion is met.
2. COBA completed eighty epochs and selected epoch 75 at 43.87%. The contemporaneous PING–COBA difference is +4.17 points.
3. Both cells have finite 80-epoch records, zero skipped/non-finite updates, matched exp068 split hashes, active populations, and matched validation raster inputs. The official SHD test was not accessed.
4. Both pods self-terminated; collection verified zero active pods. Exact billing reconciliation remains pending before publication.



## Visible messages added in CP-005

### Assistant

2026-07-19 01:44:46.210 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Dispatch checkpoint **9b5007c** is pushed and PR 52 has timestamped smoke/compute comments. A small logging defect is visible in the second PR comment: its commit placeholder was posted literally because shell interpolation was intentionally suppressed; I'll append a correction with **9b5007c** rather than edit history. The scientific run is unaffected.

### Assistant

2026-07-19 01:46:01.047 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain active after provisioning and have not self-terminated early, so there is no immediate kill-condition signal. Current accrued exposure is still well below \$1; I'll continue monitoring for independent completion and reap immediately if either backstop or failure state appears.

### Assistant

2026-07-19 01:47:15.819 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Exp069 is actively running.

- Smoke passed for both COBA and PING.
- Implementation/smoke commit: **b944ebf**
- Dispatch checkpoint: **9b5007c**
- Exactly two pods active, one per architecture.
- Rate: \$0.99/hour each; worst-case exposure \$7.92.
- Current spend remains below \$1.
- Official SHD test remains inaccessible.
- PR 52 has timestamped smoke and dispatch logs.

Next: wait for completion, collect results, reap all pods, publish artifacts/reports, and finish at the review gate.

### Assistant

2026-07-19 01:47:21.979 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm resuming from the live dispatch checkpoint and checking the authoritative fleet/worktree state now. If the cells have finished, I'll collect and reap immediately; otherwise I'll continue bounded monitoring against the \$10 ceiling.

### Assistant

2026-07-19 01:47:31.663 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The worktree is clean and both registered cells are still running normally. At roughly four minutes after dispatch, accrued exposure is about \$0.13, leaving ample budget and backstop margin. I'll monitor at sub-minute intervals; no duplicate compute will be started.

## Assistant

2026-07-19 01:48:43.449 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both exp069 cells are still running normally.

- Active pods: 2
- Duplicate jobs: 0
- Worktree: clean
- Estimated accrued spend: approximately \$0.17
- Hard ceiling: \$10
- Four-hour backstops remain active

Monitoring continues; collection, reaping, publication, and final validation remain pending.

## Assistant

2026-07-19 01:48:49.611 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm checking the existing monitor and provider state without starting anything new. If one cell has completed, I'll keep the other under its registered backstop; when both are gone, collection and exact spend accounting start immediately.

## Assistant

2026-07-19 01:49:03.227 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain active about six minutes after dispatch, with estimated combined exposure around \$0.20 and no early termination. Exp068's 40-epoch cells took roughly 68 minutes each, so an 80-epoch completion near two hours is expected; the four-hour guards still leave substantial margin.

## Assistant

2026-07-19 01:50:04.225 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain healthy and active.

- Elapsed since dispatch: about 7 minutes
- Estimated combined spend: about \$0.23
- Active pods: 2
- Worktree: clean
- No duplicate compute or protocol change

Expected runtime is roughly two hours based on exp068. Monitoring remains active under the four-hour/\$10 safeguards.

**Assistant**

2026-07-19 01:50:09.066 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm querying the persistent monitor rather than creating another polling process. The current stage is expected waiting, not a blocker; safeguards remain the controlling condition.

**Assistant**

2026-07-19 01:50:52.418 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both registered cells remain active at 01:50:46 UTC.

- Elapsed: about 8 minutes
- Estimated combined spend: about \$0.27
- Active pods: 2
- Budget/backstops: intact
- No duplicate jobs or protocol changes

Collection and publication remain pending completion.

**Assistant**

2026-07-19 01:51:05.190 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

While compute runs, I'm using the wait productively: auditing the public activity trace against the immutable transcript. CP-003 currently compresses one long visible handoff prompt, which violates the explicit verbatim requirement; I'm correcting it now, with only the required private-path redaction. This does not touch the experiment.

**Assistant**

2026-07-19 01:52:31.076 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The trace audit is corrected and pushed at `ccb8ad5`; the long handoff is now verbatim except for its explicitly marked private-path redaction, and the 37-entry build passes. Both compute cells remain active. I'm drafting the artifact-driven report structure next, leaving every result sourced from `numbers.json` once collection completes.

**Assistant**

2026-07-19 01:52:44.832 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Progress during compute wait:

- Corrected CP-003 to preserve the full visible handoff prompt verbatim, except its explicitly marked private-path redaction.
- Rebuilt all 37 Demolab entries successfully.
- Committed and pushed the correction as `ccb8ad5`.

- Worktree is clean.
- Both registered cells remain active.
- Estimated combined spend at 01:52 UTC: about \$0.33.

Collection, reaping, results, and publication remain pending cell completion.

### Assistant

2026-07-19 01:52:49.984 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm checking whether either self-terminating cell has completed. No new compute will be created; the next mutation remains collection/reaping only after the fleet changes.

### Assistant

2026-07-19 01:53:56.517 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain active at 01:53:51 UTC.

- Elapsed: about 11 minutes
- Estimated combined spend: about \$0.36
- Active pods: 2
- Worktree: clean
- Backstops and budget ceiling remain intact

No intervention is warranted yet; collection waits for self-termination.

### Assistant

2026-07-19 01:54:02.222 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll make one more bounded state check this cycle. The cells are still far inside their expected runtime, so absence of self-termination alone is not an anomaly.

### Assistant

2026-07-19 01:55:04.149 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain active at 01:54:59 UTC.

- Elapsed: about 12 minutes
- Estimated combined spend: about \$0.40
- Active pods: 2
- No anomaly or intervention trigger
- Budget and four-hour backstops remain intact

Awaiting registered completion.

**Assistant**

2026-07-19 01:55:17.145 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The run remains in its expected long-training window. I'm maintaining the existing two-pod fleet and will not infer scientific progress from billing state alone; only collected artifacts will determine success or failure.

**Assistant**

2026-07-19 01:56:18.894 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain active at 01:56:13 UTC.

- Elapsed: about 13 minutes
- Estimated combined spend: about \$0.44
- Active pods: 2
- Budget/backstops remain intact
- No protocol intervention or duplicate compute

Awaiting artifact-backed completion.

**Assistant**

2026-07-19 01:56:25.941 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The scientific state is unchanged; I'm continuing bounded monitoring. Exact spend will be calculated from observed creation-to-termination intervals, not these interim estimates.

**Assistant**

2026-07-19 01:57:25.961 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain active at 01:57:21 UTC.

- Elapsed: about 14 minutes
- Estimated combined spend: about \$0.47
- Active pods: 2
- Safeguards remain intact
- No scientific conclusion is being drawn before artifact collection

Awaiting completion.

**Assistant**

2026-07-19 01:57:29.998 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm continuing the registered wait. Since the expected duration is hours rather than minutes, interim fleet checks are only safety checks; they are not scientific evidence.

**Assistant**

2026-07-19 01:58:32.871 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells remain active at 01:58:26 UTC.

- Elapsed: about 15 minutes
- Estimated combined spend: about \$0.51
- Active pods: 2
- No safety or protocol trigger
- Budget/backstops remain intact

Awaiting artifact-backed completion.

**Assistant**

2026-07-19 01:58:43.328 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'm switching to a persistent bounded monitor so the same safe state is not re-polled manually each continuation. It will report only sanitized fleet counts, stop when zero pods remain, and never create or modify compute.

**Assistant**

2026-07-19 01:58:52.619 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

A persistent sanitized monitor is now active.

- Active pods at start: 2
- Poll interval: 50 seconds
- Maximum monitoring window: bounded beyond the four-hour pod backstops
- Mutations: none
- It will stop automatically when the fleet reaches zero

The next continuation can collect immediately once the monitor reports completion.

**Assistant**

2026-07-19 01:59:00.019 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The persistent monitor is already running, so I'm reading its next update rather than issuing another provider poll.

**Assistant**

2026-07-19 02:01:12.635 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The monitor confirms two active cells on consecutive checks. No fleet transition or safety trigger has occurred; I'm continuing the same monitor.

**Assistant**

2026-07-19 02:13:29.973 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Ten minutes of sanitized monitoring show the fleet stable at two. This is still within expected training time and well inside the budget envelope; I'm continuing without intervention.

**Assistant**

2026-07-19 02:33:38.476 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The fleet remains at two through roughly the first 45 minutes, matching the expected long epoch loop rather than an early failure. Estimated combined exposure is still under \$1.50.

**Assistant**

2026-07-19 02:53:50.129 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At roughly one hour after dispatch, both cells are still active. Exp068 needed about 68 minutes for 40 epochs, so this remains consistent with the registered 80-epoch workload. No action is indicated.

**Assistant**

2026-07-19 03:24:01.077 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The fleet remains stable through roughly 95 minutes, with estimated combined exposure around \$3.1. That is still below half the \$10 ceiling and before the expected doubled-duration completion window.

**Assistant**

2026-07-19 03:54:13.473 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells are still active near the two-hour mark. This is close to the linear runtime estimate from exp068 but not beyond the registered four-hour limit; I'm continuing without altering the run.

**Assistant**

2026-07-19 04:04:19.610 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

One cell has now self-terminated successfully enough to leave the fleet; one remains active under its original backstop. I'm not collecting early because the shared artifact set is incomplete, and I'm not inferring which scientific criterion passed until both raw outputs are collected and validated.

**Assistant**

2026-07-19 04:08:51.267 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both cells have now self-terminated and the active fleet is zero. I'm beginning collection immediately, then I'll verify the raw completion/failure markers before constructing the exact cost ledger or publishing any result.

### Assistant

2026-07-19 04:09:28.363 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Collection contains both complete markers and no failure files. The primary result clears the registered threshold: PING selected 48.04% at epoch 79, a +4.54-point gain over 43.50%; COBA selected 43.87% at epoch 75. Both split hashes match exp068. I'm now validating all 80 epoch records and deriving the exact observed cost from lifecycle timestamps before publication.

### Assistant

2026-07-19 04:12:10.443 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The integrity gate passes: both cells have exactly 80 finite epoch records, zero skipped/non-finite updates, active selected populations, matched validation raster inputs, and zero active pods. I found one publication-only omission before generating artifacts—the registered diagnostics require COBA's change from 40.81% and the contemporaneous PING-COBA gap in `numbers.json`. I'm adding those derived fields without changing any run or criterion.

## Checkpoint CP-006

This publication checkpoint follows result commit `5e80b23`. Its immutable private source has SHA-256 prefix `afa0dba76b1b`, and its checkpoint time is `2026-07-19 05:05:12.446 UTC`.

### Publication ledger

1. Delayed provider billing was allowed to settle before publication, avoiding a material spend understatement. Authoritative charges total 4.627566222 USD; zero pods remain.
2. The artifact bundle contains provenance, sixteen verified raw hashes, reproducer scripts, learning/activity curves, diagnostics, and matched validation rasters. Visual raster inspection passed.
3. Two report builds exposed punctuation-adjacent Typst interpolation errors. Both were presentation-only, were corrected without changing results, and the final 38-entry build passed with zero stubs.
4. The result commit was pushed. Focused Ruff, type checking, compilation, registered assertions, epoch integrity, raw hashes, matched inputs, reproducer syntax, official-test-route absence, zero-pod state, and remote tracking all passed.
5. The normal PR metadata edit hit a GitHub Projects-classic deprecation error. The authorized REST fallback updated PR 52 successfully; it remains draft, open, and unmerged.



## Visible messages added in CP-006

**Assistant**

2026-07-19 04:15:18.960 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The provider's billing-history endpoint is lagging termination: it currently reports only about 91 billed minutes per pod, less than the collected 128–135 minute training runtimes, so publishing that total would understate spend. I'm waiting for the authoritative per-pod billing records to settle while keeping the fleet at zero; no further compute cost is accruing.

**Assistant**

2026-07-19 04:40:39.868 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Billing reconciliation is progressing monotonically but is still incomplete at \$4.013: that remains below the GPU-time implied by the collected runtimes plus provisioning. I'm extending the zero-cost reconciliation wait until the per-pod records cover the observed lifecycles; this does not spend additional money.

**Assistant**

2026-07-19 05:01:07.033 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The authoritative billing records have settled and now cover 8,472.906 seconds for COBA and 8,282.638 seconds for PING. Exact provider charges are \$2.340118571 and \$2.287447651, totaling \$4.627566222—well below the \$10 ceiling. I'm freezing this ledger, generating the artifact bundle, then visually checking the figures before writing the final scientific record.

**Assistant**

2026-07-19 05:02:39.801 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The artifact bundle is published with provenance, raw hashes, reproducer, both learning/activity curves, diagnostics, and matched validation rasters; visual raster inspection confirms byte-matched inputs and active PING E/I dynamics. I've now made the report and night Digest fully artifact-driven and am running the complete Demolab build.

**Assistant**

2026-07-19 05:02:45.073 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first full build found a Typst interpolation ambiguity in exp069: punctuation immediately after a JSON field was parsed as part of the key. This is report syntax only; I'm separating the rendered count from the following word and rebuilding.

**Assistant**

2026-07-19 05:04:12.822 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The complete 38-entry build now passes with zero stubs. The first report build exposed two punctuation-adjacent JSON interpolation errors; both were presentation-only and are preserved as the publication attempt before the successful rebuild. I'm committing the artifact-backed result checkpoint now, then I'll run the final integrity audit and add the closing activity checkpoint.

**Assistant**

2026-07-19 05:04:56.861 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The result checkpoint is **5e80b23**. Final integrity checks pass: all 16 raw hashes, registered result assertions, 80 finite epochs per cell, shell reproducers, official-test route absence, Ruff, type checking, compilation, zero pods, and clean remote tracking. I'm correcting the result PR comment's literal commit placeholder, replacing the stale PR description with the evidence Digest, and adding the final hash-linked activity checkpoint.

**Assistant**

2026-07-19 05:05:12.446 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The correction comment posted, but `gh pr edit` hit GitHub's deprecated Projects-classic GraphQL field before changing the PR metadata. I'm recording that external tooling failure and using GitHub's direct pull-request REST update for the same authorized title/body; the branch and experiment remain untouched.

## Checkpoint CP-007

This rendered-preview and CI checkpoint follows **4a5f8de**. Its immutable private source has SHA-256 prefix **033ebf531a5b**, and its checkpoint time is **2026-07-19 05:09:50.681 UTC**.

### Review-gate ledger

1. PR 52 has the completed Digest and remains draft, open, mergeable, and unmerged. The preview workflow passed; ar070, ar071, and exp069 each return HTTP 200 from the Pages preview.
2. The final lint workflow is red only from ten pre-existing ty diagnostics in unchanged older experiments and tests. Exp069 has no CI diagnostic and its focused type check passes; unrelated debt was left outside scope and documented on the PR.
3. The CI test job remained in progress at this checkpoint. Local execution of the repository-wide suite was intentionally omitted under the locked 4 GB host constraint.

### Visible messages added in CP-007

**Assistant**

2026-07-19 05:06:41.260 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The final activity checkpoint builds successfully across all 38 entries. It records the delayed-billing safeguard, both presentation-only build failures, the GitHub metadata fallback, the full visible transcript through publication, and private source hash `afa0dba76b1b`. I'm pushing this last checkpoint and then verifying the rendered preview/check state and requirement-by-requirement completion.

### Assistant

2026-07-19 05:07:23.862 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 52 now has the final Digest via the REST fallback and remains draft/open. The final branch checkpoint is `4a5f8de`; I'm waiting for its preview workflow, then I'll verify the three rendered pages directly and inspect any CI failure before the human-gate handoff.

### Assistant

2026-07-19 05:08:30.732 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The final lint check failed while preview and tests are still running. I'm applying the GitHub CI-fix workflow now: inspect the failing Actions log first and determine whether `exp069` introduced a defect or the failure is the documented repository-wide type debt. I won't broaden scope to unrelated files.

### Assistant

2026-07-19 05:08:36.777 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The lint failure is entirely pre-existing: 10 `ty` diagnostics in unchanged `exp041/048/054/065` and older `exp049/065` tests; `exp069` emits none, and its focused `ty` check passes. Fixing those files would exceed this night's scope, so I'm documenting the residual CI state on PR 52 and leaving code unchanged.

### Assistant

2026-07-19 05:09:50.681 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The GitHub preview is green, and all three rendered pages return HTTP 200. The only remaining external check is the already-running CI test job; I'm waiting for it so the final handoff distinguishes verified success from pending state.